

Semantic Failure Example & AI Quality Controls

REQUIRED SEMANTIC-FAILURE EXAMPLE

1. Schema-Valid but Semantically Wrong Record:

```
{
  "claim_id": "C00000088",
  "member_id": "MBR-00042",
  "procedure_code": "99285",
  "billed_amount": 1200.00,
  "allowed_amount": 1200.00,
  "date_of_service": "2026-02-15",
  "decision_status": "Approved"
}
```

2. Explanation of AI Processing & Misinterpretation:

What the AI Receives: The input payload passes structural schema validation: `allowed_amount` is a non-negative decimal (\$1,200.00) bounded by `billed_amount` (\$1,200.00), all data types match structural specifications, and non-null constraints are satisfied.

Semantic Misinterpretation: The adjudication pipeline calculates `allowed_amount` using the reimbursement rate active today (\$1,200.00) rather than evaluating the fee schedule in effect on the `date_of_service` (2026-02-15), when the approved contract rate was \$850.00. Furthermore, it ignores the member's temporal eligibility window from `patient_accounts`, treating the current "Active" status as retroactive truth even though coverage was inactive on the service date. Downstream predictive models (such as financial liability forecasting or fraud detection engines) accept this record as a valid, baseline reimbursement, distorting training features and underestimating payer liability risks.

3. Corrected / Canonical Representation:

```
{
  "claim_id": "C00000088",
  "member_id": "MBR-00042",
  "procedure_code": "99285",
  "billed_amount": 1200.00,
  "allowed_amount": 0.00,
  "date_of_service": "2026-02-15",
  "decision_status": "Denied",
  "denial_reason_code": "ERR_ELIG_POINT_IN_TIME"
}
```

AI CONTEXT / QUALITY CONTROLS

RULE / EXPECTATION	WHAT GOES WRONG	WHAT AI RECEIVES	POSSIBLE AI IMPACT	CONTROL / RESPONSE
Monetary Unit Consistency (billed_amount, allowed_amount)	Ingestion pipeline ingests source data formatted in thousands instead of single USD decimals.	billed_amount = 120 instead of 120000	Risk and utilization models severely underestimate claim cost magnitude and distort financial severity features.	Automated schema normalization check + assertion gate verifying values fall within expected standard deviation bounds.
Point-in-Time Fee Schedule Validity (allowed_amount as-of date_of_service)	Source system calculates reimbursement using current fee schedules instead of the policy rate	allowed_amount reflects current fee schedule (\$1,200.00) rather than 2026-02-15 rate (\$850.00)	Audit models falsely validate decision compliance, leading to	Point-in-time join assertion in feature engineering ETL to enforce temporal alignment with effective policy tables.

RULE / EXPECTATION	WHAT GOES WRONG	WHAT AI RECEIVES	POSSIBLE AI IMPACT	CONTROL / RESPONSE
	effective on date_of_service.		unnoticed overpayment leakage and policy drift.	
Temporal Coverage Boundary (patient_accounts Eligibility Join)	Pipeline joins member coverage using present-day eligibility_status rather than status active on date_of_service.	eligibility_status = Active (current) for a claim incurred during a historical coverage gap	Automated claims adjudication models predict approval for non- covered events, resulting in improper billing.	Temporal feature store control enforcing bi- temporal range joins (effective_start_date ≤ date_of_service ≤ effective_end_date).